

Seeking *DevOps Engineer, Platform Engineer, SRE, Cloud Infrastructure Architect* — I build and operate production infrastructure at scale, with deep expertise in CI/CD, containerization, IaC, and cloud platforms.

## Highlights

### “ Cloud Platforms & Infrastructure as Code

- Proficient in AWS (EC2, S3, Lambda, ECS, RDS, Aurora, VPC, Transit Gateway, Route53, CloudFront, API Gateway, SQS, SNS, KMS, Secrets Manager, Bedrock, OpenSearch, Athena, Glue).
- Proficient in Azure (Synapse, ADLS, Entra ID, RBAC, Power Automate).
- Infrastructure as Code: Terraform, Pulumi, Kustomize, Helm, Ansible.

### “ Container Orchestration & GitOps

- Proficient in Kubernetes (K8S), Red Hat OpenShift (ROSA), Docker, Containerd, Helm, ArgoCD, Harbor.
- Hands-on multi-cluster management with auto-scaling, drift detection, and Day 0/1/2 automation.

### “ CI/CD Pipelines & Automation

- Expert in designing and maintaining CI/CD: GitHub Actions, Jenkins, Bitbucket Pipelines, Copado, Semantic Release.
- Reusable workflow orchestration, multi-branch builds, matrix testing, A/B deployment, GitOps promotion models.

### “ Observability, Monitoring & SRE

- Proficient in Splunk, CloudWatch, Prometheus, Grafana, ELK.
- Built real-time health dashboards with alerting (SNS, WebSocket).
- Production incident management and post-mortem processes.

### “ Security & Compliance

- Security scanning: Trivy, Veracode, SonarQube, Burp Suite, Qualys.
- Secrets management: AWS Secrets Manager, Hashicorp Vault, Ansible Vault.
- FedRAMP SSP (NIST 800-53), SBOM compliance, CVE remediation.
- SSO/Identity: OAuth2, OIDC, SAML, SCIM, Azure AD, Keycloak.

### “ Scripting & Development

- Python, Bash, Groovy, YAML, Terraform HCL, Ansible playbooks.
- Django, FastAPI, Flask for tooling and automation services.
- Strong in Linux, networking (BGP, Direct Connect, VPN, WAF).

## Experience

**12/2023** Department of Health and Human Service, North Carolina, Principal System Architect / DevOps Lead

- “ Managed multi-cluster ROSA (Red Hat OpenShift on AWS) infrastructure (Dev, Pre-Prod, Prod) with auto-scaling worker nodes (r6a.2xl, up to 12 nodes/96 CPU/768 GB) across 7 AWS accounts and 3 Azure subscriptions. Implemented Terraform-based IaC, Kustomize manifests, and ArgoCD GitOps with drift detection and Day 0/1/2 automation.

- “ Designed and maintained centralized CI/CD pipeline orchestration via reusable GitHub Actions workflows shared across 50+ repositories, integrating MuleSoft deployment, SonarQube quality gates, Postman API testing, security scanning (Trivy, Veracode), Semantic Release, and commitlint enforcement.
- “ Built a real-time PROD health monitoring system checking 48+ API endpoints and OpenShift pods every 60 seconds with AWS SNS alerting on consecutive failures, React UI with WebSocket updates, complementing Splunk and CloudWatch dashboards.
- “ Architected serverless infrastructure for a document management system (replacing IBM FileNet): event-driven pipeline (S3, EventBridge, Lambda, DynamoDB, SQS DLQ), KMS encryption, VPC Endpoints, JWT custom authorizer, automated Glacier archival, deployed via Terraform/OpenTofu across dev/test/prod.
- “ Managed Azure infrastructure: Synapse Workspace, Dedicated SQL Pool, ADLS, Entra ID/RBAC, Power Automate. Architected end-to-end data pipeline (Salesforce → Azure Pipeline → Synapse → Power BI → AWS SFTP → NCFS mainframe) for monthly financial batch processing.
- “ Deployed GenAI platform on OpenShift via ArgoCD: Django/FastAPI backend with Celery/Redis workers, AWS Bedrock integration, OpenSearch Serverless, Terraform modules for Lambda/S3/RDS across 6 environments with Docker Compose for local dev.
- “ Implemented Terraform-based AWS Secrets Manager automation with per-secret IAM resource policies, metadata tagging, rotation alerting via GitHub Actions, and certificate lifecycle management.
- “ Designed the user identity pipeline: Azure AD/NCID SSO with SCIM provisioning to Salesforce, JIT activation, automated user validation tooling (Lambda + S3 + API Gateway).
- “ Led security hardening: FedRAMP SSP documentation (NIST 800-53), SBOM and license compliance tracking for 30+ SaaS products, network security across Direct Connect, Transit Gateway, Cloudflare WAF, and security group automation.
- “ Built high-performance ETL service (Python/FastAPI) as OpenShift CronJob: monthly SFTP ingestion of 1.2M+ records, streaming delta loads to PostgreSQL, Athena serverless analytics, secured via MuleSoft OAuth.
- “ Initiated comprehensive test automation infrastructure: Robot Framework regression suites, Playwright E2E, JMeter load testing, YAML-driven declarative test framework, all integrated into CI/CD.

**3/2022 - 3/2023** - Lucidum, Principal System Architect

- “ As primary developer and maintainer of CI/CD pipelines featuring multi-branch building, matrix testing, A/B deployment, integration of GitHub Code, GitHub Actions, GitHub Package, AWS, and Azure.
- “ Managed successful execution of 2 major releases, 6 minor releases, and 12 patch releases. Initiated GitOps practices using GitHub Actions, GitHub Package Registry, and Semantic Release.
- “ Hardened security in DevOps workflows by initiating CVE scanner (code, docker image), and integration with secret managers including Hashicorp Vault, Ansible Vault, AWS & Azure Secret Managers.

- “ As the company’s AWS and Azure administrator responsible for overall configuration and application of cloud services including SSO, IAM, ACR, ECR, ECS, RDS, Aurora, EC2, S3, Fargate, Lambda, CloudWatch, SNS, VPC, WAF, ELB, TargetGroup, Route53, CloudFront.
- “ Transformed on-prem-only deployment model to hybrid and cloud-native. Introduced and integrated various AWS & Azure services to achieve a successful transition of twelve production systems.
- “ Drove substantial improvements in test automation: revamped 800+ Cypress E2E tests within 6 weeks, improved performance by 50% and achieved 100% automation in UAT.

**11/2016 - 3/2022** Lenovo US, P8, Advisory Engineer, Senior Solution Architect, Team Lead

- “ As the principal system architect and first author of the *“Lenovo Open Cloud Automation Reference Architecture”*, a baremetal-to-cloud orchestration platform. Led a global team of six, delivered in 13 months, showcased at MWC Barcelona 2019, won \$100M contract. Key DevOps: Jenkins, Ansible, Python, zero-touch provisioning.
- “ As the principal system architect of the *“RedHat Hyperconverged Infrastructure (RHHI) Reference Architecture”*, a 3-12 server VM-workload solution with Glusterfs, layer-3 networking, HA, fault tolerance, zero-touch node discovery. Key DevOps: Ansible, Jenkins, Cloudform, Python.
- “ As the principal architect of the *“Lenovo Workload Solution Store”*, a baremetal orchestration solution using declarative taxonomy, serving as the core of ThinkAgile VX. Key DevOps: Ansible, Canonical MAAS, Juju, Django, React.
- “ As team lead of ThinkAgile CP, a hybrid cloud platform with on-prem infrastructure and AWS management plane. Oversaw DevOps in Jenkins, ArgoCD, Helm, Kubernetes (K8S), and production deployment in AWS.
- “ As lead developer of a DCIM system (Netbox-based) administrating 300+ servers and switches globally. Stack: Django, Celery, Redis, MySQL, Docker, Ansible, ENOS/NOS.

**01/2015 - 11/2016** PY Consulting, Founder, Freelance

*UNC Chapel Hill, School of Medicine, Contractor*

- “ Initiated the organization’s first Docker-based deployment and CI/CD in production. Led the school’s DevOps initiative by authoring its technology roadmap, and oversaw DevOps Phase 2 & Phase 3. Provided trainings in Git, Docker, Jenkins, Python, Ansible, Cypress.

*Massachusetts General Hospital, Volunteer*

- “ Containerized deployment of two ML tools (deeplabcut, simba), reducing installation time from days to 30 minutes. Mentored team on Git, Docker, Python, DevOps.

**04/2004 - 05/2007** Bit 9 Inc. (Carbon Black), Researcher, Tech Lead

- “ Led research securing the Windows NT kernel, resulting in \$6M A-round from Kleiner Perkins.

**07/1998** Instron Corp., Senior Engineer

**- 04/2004**

- “ Developed real-time firmware: VxWorks RTOS, real-time Linux, C, National Instrument DAQ, Texas Instruments DSP, Allen-Bradley PLC.

## Education

- “ *MBA (04/2007 - 09/2008)*, Boston Univ., International Management & Finance
- “ *Master of Science (09/1995 - 05/1997)*, Ohio Univ., Electrical Engineering
- “ *Bachelor of Science (09/1990 - 07/1994)*, Shanghai JiaoTong Univ., China, Electrical Engineering

## Publications

1. <https://lenovopress.com/lp1148-red-hat-hyperconverged-infrastructure-for-virtualization-reference-architecture>
2. <https://lenovopress.com/lp1149-lenovo-open-cloud-reference-architecture>